

Exam Date & Time: 06-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Distributed Cloud Security [CSE-4015]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Identify some real-world applications of distributed systems, and how do they benefit from distributed architecture. (5)
 - A)
 - B) Analyse the advantages and disadvantages of biometric authentication compared to other authentication methods. (3)
 - C) In a corporate network environment, a company's IT department manages access to various network resources, such as files and folders stored on a central file server. The IT department utilizes access control lists (ACLs) to regulate and restrict access to these resources based on user permissions. Examine the process of creating the access control lists (ACLs), and explain how they are used to manage access to resources? (2)
- 2) List some common implementation challenges associated with Discretionary Access Control. (5)
 - A)
 - B) Identify some common vulnerabilities found at the application layer, and how can developers mitigate these risks? (3)
 - C) Explain about implementation of protection domains using access control lists. (2)
- 3) Identify any two different scenarios for illustrating SQL injection attack. (5)
 - A)
 - B) Consider a scenario in the future where security measures against Application-Based Bandwidth attacks are extensively adopted compared to the current situation. Despite the advancements, should server system administrators remain vigilant and implement additional countermeasures against Application-Based Bandwidth attacks? If affirmative, what types of attacks could persist, and what strategies can be employed to mitigate their impact? (3)
 - C) Explain about essential characteristics architectural influences involved in cloud computing. (2)

- 4) A multinational corporation decides to migrate its critical business applications and data to a public cloud provider to leverage scalability and cost-efficiency. However, during the migration process, a misconfiguration in the cloud infrastructure allows unauthorized access to the company's sensitive data. Assuming you are a cloud administrator. Then, what will you consider as fundamental principles of cloud security design? (5)
- A)
- B) To execute a DNS amplification attack, the attacker needs to induce the generation of a significant volume of DNS response packets from the intermediary to overwhelm the capacity of the link to the target organization. Let's consider an attack scenario where the DNS response packets are 1200 bytes in size. How many of these packets per second should the attacker trigger to flood a target organization using links with different capacities: 1 Mbps, 4 Mbps, or 16 Mbps? Additionally, if the DNS request packet to the intermediary is 80 bytes, what amount of bandwidth does the attacker consume to transmit the required rate of DNS request packets for each of these three cases? (3)
- C) Compare and contrast different cloud delivery models. (2)
- 5) Analyse the unique challenges of testing software deployed in cloud environments compared to traditional on-premises environments. (5)
- A)
- B) Identify the list of considerations should organizations take into account when selecting disaster recovery plan services for their business continuity needs? (3)
- C) Analyse the significant factors organizations should consider when deciding which cloud delivery model to adopt for their applications and services. (2)

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